

Supplementary Material of "MCIQA-2K: A Multi-Dimensional Dataset and No-Reference Quality Assessment Benchmark for Colorized Images"

1 Details of MCIQA-2K dataset

We curate 400 high-quality grayscale images from the COCO test2017 dataset[9]. We adopt five mainstream colorization models with distinct network architectures: CIC[16](CNN), GCP-Colorization[15](GAN), BigColor[8](GAN), CT²[14](Transformer) and DDColor[6](Transformer). In total, we obtain 2000 colorized images.

Table 1. Performance of different metrics on the MCIQA-2K dataset based on the 5 different colorization models. The reference images of all FR metrics are the original RGB images from COCO test set[9]. $\uparrow$ means the bigger value is the better. $\downarrow$ means the smaller value is the better. The value of each IQA on each dataset is the average result of 3 experiments with different random seeds.

Metric	BigColor	CIC	CT ²	DDColor	GCP
PSNR $\uparrow$	19.9407	21.6503	21.5569	22.5734	20.7946
SSIM[13] $\uparrow$	0.9283	0.9943	0.9903	0.9941	0.9955
LPIPS[17] $\downarrow$	0.2071	0.1882	0.1708	0.1437	0.2029
DISTS[4] $\downarrow$	0.1209	0.1171	0.1202	0.0783	0.1114
BRISQUE[10] $\downarrow$	18.8400	12.9696	13.2542	12.7498	12.8829
NIQE[11] $\downarrow$	3.2440	3.6785	3.7036	3.6912	3.6812
CNN-IQA[5] $\uparrow$	0.6989	0.6968	0.6932	0.7021	0.6979
DBCNN[18] $\uparrow$	0.6596	0.6695	0.6711	0.6860	0.6733
WaDIQaM[2] $\uparrow$	0.1112	0.0997	0.1017	0.0876	0.0540
MUSIQ[7] $\uparrow$	71.7160	70.0860	70.2506	71.6514	71.0610
CLIP-IQA[12] $\uparrow$	0.5545	0.5329	0.6092	0.5961	0.5638
CLIP-IQA+[12] $\uparrow$	0.6733	0.6644	0.6742	0.6775	0.6590
QualiClip[1] $\uparrow$	0.7532	0.6679	0.7409	0.7664	0.7284

The quantitative statistics in Table 1 further demonstrate the substantial quality differences among the selected colorization models, validating the diversity and representativeness of the proposed dataset. Specifically, DDColor[6] consistently achieves the best performance across most full-reference and learning-based perceptual metrics, including PSNR, LPIPS[17], DISTS[4], DBCNN[18] and QualiClip[1], indicating its superior capability in generating semantically

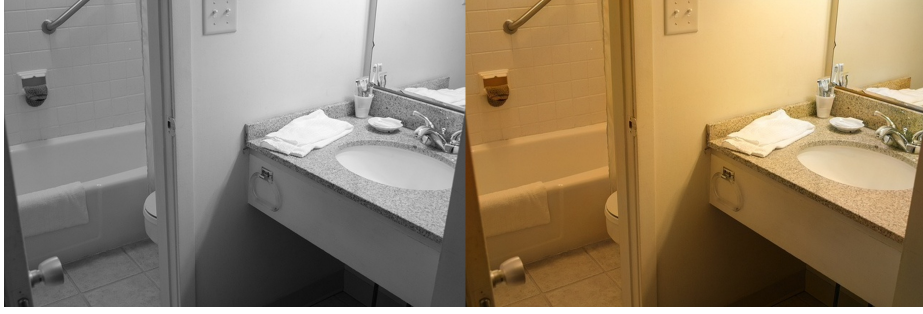


Fig. 1. Example image samples provided to participants in the crowdsourcing study.

plausible and visually natural colorization results. In contrast, BigColor[8] exhibits relatively lower PSNR and higher perceptual distances, suggesting that its generated images contain more noticeable color inconsistencies and structural deviations despite producing vivid colors. Meanwhile, CT²[14] and CIC[16] achieve competitive structural similarity scores but still demonstrate observable gaps in perceptual-oriented metrics such as CLIP-IQA and QualiClip, reflecting differences in semantic realism and perceptual naturalness. GCP-Colorization[15] produces relatively balanced results but remains inferior to DDColor[6] on most evaluation criteria. Overall, these significant discrepancies across different IQA metrics confirm that the selected colorization models generate diverse visual characteristics and perceptual artifacts, thereby ensuring that MCIQA-2K sufficiently captures a wide spectrum of colorization quality variations and distortion patterns encountered in practical scenarios.

2 Details of Human Annotation

All subjective evaluation procedures strictly followed the standard guidelines specified in [3]. All test images were presented on a 27-inch Dell S2721DS IPS monitor featuring a native resolution of 2560×1440 and a 75 Hz refresh rate. The monitor was professionally calibrated to achieve stable luminance and accurate color rendition, with fixed brightness and contrast configurations throughout all evaluation sessions and all dynamic image enhancement functions disabled. All participants maintained a standardized viewing distance equivalent to 2–3 times the screen height to guarantee identical viewing angles. The experimental environment adopted dim and uniform ambient illumination, eliminating direct light irradiation and screen reflections. A neutral surrounding background was adopted to prevent potential visual preference bias, and all assessments were carried out in a quiet space to reduce external interference. Stimulus images were displayed in full-screen mode with image scaling deactivated, so as to retain original spatial resolution and eliminate unwanted interpolation distortions. Each rater independently assessed three-dimensional perceptual visual quality of colorized images following a standard 1-to-5 scoring scale.

Notably, to exclude full-reference bias, raters were given only the paired grayscale version converted from each colorized image for reference, which was positioned to the left of the colored output. Each colorized image was displayed individually during scoring, as visualized in Fig. 1. Such a design drives subjective judgments to focus on perceptual realism rather than strict alignment with a single reference standard.

3 Ablation Study

To further investigate the contribution of the proposed CS and SCM branches to GN prediction, we conduct an ablation study on the MCIQA-2K dataset. Specifically, we evaluate the performance of MCIQA-GN under three settings: the complete model, the model without the CS branch, and the model without the SCM branch. All variants are trained under identical settings and training epochs to ensure a fair comparison. The quantitative results are reported in Table 2.

Table 2. Performance comparison of MCIQA-GN with disabling specific branches on MCIQA-2K dataset. 3 models are trained based on the same conditions and epoches.

Model	SRCC $\uparrow$	PLCC $\uparrow$
MCIQA-GN	0.8511	0.8571
without CS	0.6281	0.6597
without SCM	0.7013	0.7434

The complete MCIQA-GN achieves the best performance with an SRCC of 0.8511 and a PLCC of 0.8571, demonstrating the effectiveness of jointly incorporating both CS and SCM information into global naturalness assessment. When the CS branch is removed, the performance drops significantly to 0.6281/0.6597 in terms of SRCC/PLCC, indicating that local chromatic consistency and color smoothness cues are highly important for evaluating the overall perceptual realism of colorized images. Similarly, removing the SCM branch also results in a noticeable performance decline, reducing the SRCC and PLCC to 0.7013 and 0.7434, respectively. This observation demonstrates that semantic-level color alignment plays a critical role in global quality perception.

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